Kadin El Bakkouri

University of Florida | B.S. Biomedical Engineering (Honors), Expected May 2027
Minors: Chemistry, Bioinformatics | GPA: 3.76

kelbakkouri@ufl.edu | elbakkouri.kadin@gmail.com | github.com/kadinelbak | linkedin.com/in/kadin-el-bakkouri-09531b289

EDUCATION

University of Florida | Bachelor of Science in Biomedical Engineering (Honors)

Expected May 2027
Minors: Chemistry, Bioinformatics
Selected Coursework: Microbiology, Biochemistry, Instrumentation, Cell Engineering Lab, Differential Equations, Physiology, Biomaterials, Quantitative Physiology, Biothermodynamics, Biotransport

RESEARCH EXPERIENCE

B.E.A.T. Cancer Lab, University of Florida | Mathematical Modeler and Research Assistant

2024-Present

- Developed differential equation models to characterize ovarian cancer population growth dynamics in sensitive and resistant cell lines.
- Built reproducible fitting workflows for monoculture and coculture systems using parameter estimation, model comparison, and information-criterion analysis.
- Investigated adaptive therapy questions by modeling ecological competition between treatment-sensitive and treatment-resistant populations.
- Automated image-analysis and modeling workflows, substantially reducing manual processing burden.
- Presented research at multiple symposiums and received 3rd Place for research poster presentation.
- Contributed to a pending manuscript examining adaptive therapy performance relative to maximum tolerated dose.
- This work forms the basis of my senior thesis.

Disease and Treatment Modeling in Ovarian Cancer

- Built interpretable ODE-based models of ovarian cancer population dynamics under untreated and treated conditions.
- Compared logistic, theta-logistic, and coculture competition models to estimate growth, carrying-capacity, and interaction parameters.
- Structured the modeling pipeline to support future treated coculture analysis relevant to adaptive therapy design.
- Generated figure-ready outputs and organized model-selection workflows using SSE, AIC, and BIC.

QuPath Cell Count Pipeline

- Built and refined a reproducible histology analysis pipeline for large-batch image ingestion, cell detection, quality-control review, and export.
- Standardized folder ingestion, naming, and batch execution to reduce inconsistency associated with manual counting.
- Improved the path from raw image folders to analysis-ready quantitative outputs for downstream statistical use.

LEADERSHIP AND SERVICE

Generational Relief in Prosthetics (GRiP) | Vice President, September 2023-Present

- Oversaw project intake, approval, and operational continuity for an organization supporting approximately 45 assistive-device projects.
- Built and maintained systems for onboarding, team assignment, project tracking, and leadership transition.

- Worked directly with families, clinicians, occupational therapists, engineers, and student teams to preserve safety, usability, and accountability.
- Helped scale organizational infrastructure while maintaining delivery boundaries and continuity across semesters.
- Mentored junior members and helped transition multiple students into leadership roles.

GRiP Designathon | Organizer and Lead

- Designed and executed a recipient-focused assistive technology designathon with 45 participants and 15 submissions.
- Built the event workflow from case framing and logistics through judging and follow-up.
- Helped convert student design work into a repeatable pipeline for future recipient-facing projects.

HealthHacks | Organizer and Lead

- Coordinated a cross-university health-tech prototyping event involving UF and USF participants.
- Managed hardware, logistics, judging coordination, and sprint execution structure.
- Supported practical prototype development under real time and resource constraints.

Captain | Team Lead

- Lead 8 members in developing a device prosthetic for a 6 year old to ride a bike with a missing forearm.
- Assigned small project tasks to pairs over the course of a year and consolidated their work into a singular model.
- Used Solidworks to create a backup model in case any member was late on a deliverable so that the next member would not be behind.

TEACHING AND MENTORSHIP

Department of Biomedical Engineering, University of Florida | Learning Assistant, August 2025-Present

- Led review sessions and supported students in analytical problem-solving.
- Completed pedagogy training focused on neuroscience-informed learning strategies.
- Helped students move from formula memorization toward mechanistic reasoning.

Center for Precollegiate Education and Training | Research Counselor, May 2024-August 2025

- Mentored students through full research cycles from idea development to presentation.
- Led laboratory activities involving ELISA, PCR, and cell culture.
- Supported medical records and program documentation for large student cohorts.

Varsity Tutors | Research Counselor, Jan 2024 - December 2024

- Mentored students through SAT test taking strategies
- Identified critical gaps in SAT knowledge to encourage practice in that specific area.
- Aided 8 students in getting college scholarships over the year.

CLINICAL EXPOSURE AND VOLUNTEERING

Shands Hospital | Volunteer, September 2023-December 2024

- Provided companionship to patients in ICU settings.
- Supported Alzheimer’s care activities through structured engagement.
- Assisted physical therapy staff while maintaining patient comfort and safety.
- Completed approximately 100 hours of hospital volunteering.

ENGINEERING AND COMPUTATIONAL PROJECTS

HIV-1 Protease Detector

- Built a drug discovery screening workflow that converts SMILES strings into molecular fingerprints and predicts HIV-1 protease inhibition.
- Trained a PyTorch model on more than 41,000 compounds.
- Integrated cheminformatics and machine learning methods using RDKit and Python.

GRiP Admin Platform

- Designed internal infrastructure for approvals, attendance history, role-aware workflows, and organizational continuity.
- Prioritized clarity, auditability, and long-term maintainability over technical novelty.
- Supported project scaling without losing accountability.

VitalIntel

- Developed a heart-disease risk modeling workflow using preprocessing, exploratory analysis, feature evaluation, and classifier comparison.
- Applied machine learning methods to clinically relevant structured data.

Heart Smart

- Designed an interactive CHF patient education device to visualize cardiac workload and fluid pooling.
- Emphasized patient communication, tangible demonstration, and human factors in device design.
- Received First Place in the Dream Team Engineering Hackathon.

GRiP Bike Prosthetics

- Developed recipient-specific cycling prosthetics through iterative CAD, fabrication planning, and team mentorship.
- Worked within real fit, comfort, and usability constraints.

Assistive and Prosthetics Design

- Led CAD work for post-operative footwear and prosthetic concepts shaped by gait constraints, stability, and material tradeoffs.
- Focused on manufacturable geometry and practical patient use.

NeuroVac

- Developed a non-invasive device concept for sinking skin flap syndrome centered on sensing, portability, and clinical safety constraints.

HONORS AND AWARDS

- 3rd Place, Research Poster Presentation
- First Place, Dream Team Engineering Hackathon
- University of Florida Honors Biomedical Engineering

TECHNICAL SKILLS

Programming

Python, Julia, MATLAB, JavaScript

Modeling and Data

Differential equations, dynamical systems modeling, scientific computing, machine learning, model fitting, quantitative analysis

Engineering Tools

SolidWorks, Fusion 360, OnShape, KiCad, OpenSim

Laboratory and Analysis

Image analysis, histology pipeline workflows, ELISA, PCR, cell culture

Research Interests

Adaptive therapy, tumor ecology, translational modeling, computational oncology, biomedical device design, health-tech systems, computational biology